

Bitwise differential trail through **SERPENT** found by SAT

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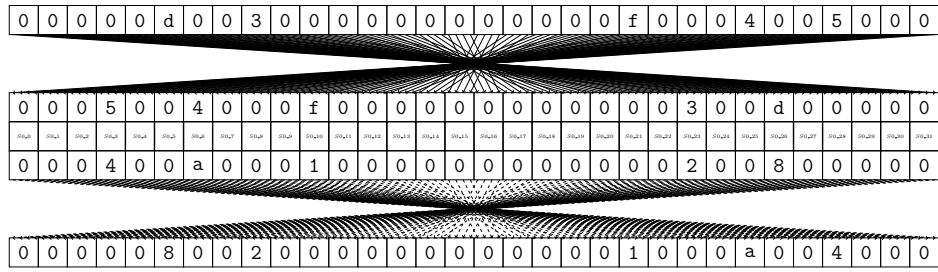
1 SERPENT

Maximal differential probability: 2^{-19}

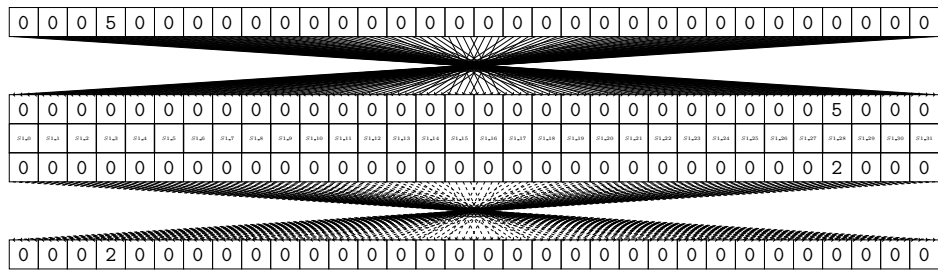
0	4	0	0	0	4	0	0	0	4	0	0	0	4	4	8	0	0	8	0	0	4	0	0	0	4	8	0	0	4	0	8	
R1																																
0	4	0	0	0	4	0	0	0	4	0	0	0	4	4	8	0	0	8	0	0	4	0	0	0	4	8	0	0	4	0	8	
R2																																
0	0	0	0	0	d	0	0	3	0	0	0	0	0	0	0	0	0	0	0	0	f	0	0	0	4	0	0	5	0	0	0	
R3																																
0	0	0	0	0	d	0	0	3	0	0	0	0	0	0	0	0	0	0	0	0	f	0	0	0	4	0	0	5	0	0	0	
R4																																
0	0	0	0	0	d	0	0	3	0	0	0	0	0	0	0	0	0	0	0	0	f	0	0	0	4	0	0	5	0	0	0	
R5																																
0	0	0	0	0	d	0	0	3	0	0	0	0	0	0	0	0	0	0	0	0	f	0	0	0	4	0	0	5	0	0	0	
R6																																
0	0	0	0	0	d	0	0	3	0	0	0	0	0	0	0	0	0	0	0	0	f	0	0	0	4	0	0	5	0	0	0	
R7																																
0	0	0	0	0	8	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	a	0	0	4	0	0	0	
R8																																
0	0	0	0	0	8	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	a	0	0	4	0	0	0	
R9																																
0	0	0	5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
R10																																
0	0	0	5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
R11																																
0	0	0	5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
R12																																
0	0	0	5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
R13																																
0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
R14																																
0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
R15																																
0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0
R16																																
0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0
R17																																
0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0
R18																																
0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0
R19																																
0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0
R20																																
0	0	f	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	e	0	0
R21																																
0	0	f	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	e	0	0
R22																																
0	0	f	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	e	0	0
R23																																
0	0	f	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	e	0	0
R24																																
2	0	0	0	0	0	0	4	2	0	0	0	0	0	0	4	2	0	0	0	0	0	0	4	2	0	0	0	0	0	0	0	0
R25																																
2	0	0	0	0	0	0	4	2	0	0	0	0	0	0	4	2	0	0	0	0	0	0	4	2	0	0	0	0	0	0	0	0

2 SBoxLayer_0

Maximal differential probability: $2^{-12.0}$



3 SBoxLayer_1

Maximal differential probability: $2^{-3.0}$ 

4 SBoxLayer_2

Maximal differential probability: $2^{-4.0}$

